

Projection of the year 2050 burden of diabetes in the US adult population: dynamic modeling of incidence, mortality, and prediabetes prevalence

Background People with diabetes can suffer from diverse complications that seriously erode quality of life. Diabetes, costing the United States more than \$174 billion per year in 2007, is expected to take an increasingly large financial toll in subsequent years. Accurate projections of diabetes burden are essential to policymakers planning for future health care needs and costs. **Methods** Using data on prediabetes and diabetes prevalence in the United States, forecasted incidence, and current US Census projections of mortality and migration, the authors constructed a series of dynamic models employing systems of difference equations to project the future burden of diabetes among US adults. A three-state model partitions the US population into no diabetes, undiagnosed diabetes, and diagnosed diabetes. A four-state model divides the state of "no diabetes" into high-risk (prediabetes) and low-risk (normal glucose) states. A five-state model incorporates an intervention designed to prevent or delay diabetes in adults at high risk. **Results** The authors project that annual diagnosed diabetes incidence (new cases) will increase from about 8 cases per 1,000 in 2008 to about 15 in 2050. Assuming low incidence and relatively high diabetes mortality, total diabetes prevalence (diagnosed and undiagnosed cases) is projected to increase from 14% in 2010 to 21% of the US adult population by 2050. However, if recent increases in diabetes incidence continue and diabetes mortality is relatively low, prevalence will increase to 33% by 2050. A middle-ground scenario projects a prevalence of 25% to 28% by 2050. Intervention can reduce, but not eliminate, increases in diabetes prevalence. **Conclusions** These projected increases are largely attributable to the aging of the US population, increasing numbers of members of higher-risk minority groups in the population, and people with diabetes living longer. Effective strategies will need to be undertaken to moderate the impact of these factors on national diabetes burden. Our analysis suggests that widespread implementation of reasonably effective preventive interventions focused on high-risk subgroups of the population can considerably reduce, but not eliminate, future increases in diabetes prevalence.